

Machine learning for predicting used car resale prices using granular vehicle equipment information

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ABSTRACT

Millions of used cars are sold every year, and, hence, accurate estimates of resale values are needed. One reason is that under- and overestimating the value of used cars at the end of their leasing period is directly related to the financial return of car retailers. However, in previous literature, granular vehicle equipment information (e.g., alloy rims, park assistance systems) as a predictor has been largely overlooked. In order to address this research gap, we assess the predictive power of granular information about vehicle equipment when forecasting the resale value of used cars. To achieve this, we first preprocess 50,000 equipment options through a tailored, end-to-end automated procedure. Subsequently, we employ machine learning using a comprehensive real-world dataset comprising 92,239 sales where each vehicle is characterized by a unique equipment configuration. We find that including equipment information improves the prediction performance (i.e., mean absolute error) by 3.27% and at a statistically significant level. Altogether, car retailers can use information about the specific vehicle configuration to more accurately predict prices of used vehicles, and, as an implication for businesses, this may eventually increase returns.

1. Introduction

In 2021, 32.7 million second-hand cars were sold in Europe (Frost & Sullivan, 2022). For each car, the resale value depends on several factors, including the specific characteristics of the vehicle, such as its make, model, mileage, and overall condition. Given the inherent volatility and opacity of the used car market (Akerlof, 1970; Bauer et al., 2020), accurately predicting resale values is a challenging task, which often involves large uncertainty for both buyers and sellers. Altogether, this highlights the need for data-driven models that predict a vehicle's market value.

Predicting the resale value of used cars becomes particularly important in the context of car leasing (Jerenz, 2008). Here, the expected resale value of the car at the end of the leasing period is used to settle the buy-back price at the start of the leasing contract. This means that the resale price must be accurately predicted several years in advance. Both overestimation and underestimation of resale values have negative implications: Overestimation of the resale price causes leasing rates that do not adequately cover the vehicle's depreciation, potentially leading the retailer to sell the used vehicle at a low margin or even at a loss. Conversely, underestimation results in inflated leasing rates, which could deter potential customers and thus negatively affect sales.

In sum, incorrect predictions of residual values are a major concern in car leasing (Fabozzi, 2008; Rode et al., 2002).

In addition to pricing buy-back options, accurate predictions of resale values are also needed for operational reasons. From a financial point, accurate estimates of resale values are necessary to quantify the financial risk of deferred or defaulted leasing contracts (Grenadier, 1996; Pirotte & Vaessen, 2008). From an operational point, accurate predictions may thus allow for continuous monitoring and thus give decision-makers the ability to identify discrepancies between the negotiated pay-back price and the expected resale value throughout the duration of a leasing contract and thus to take preemptive measures. Examples of such preemptive measures are targeted offers for early buy-outs or discounts on new vehicles if the old one is traded in, making it more appealing for customers to lease another vehicle instead of buying the old one.

To predict the resale values of cars, several works have developed tailored machine learning models. Here, prior works typically use the resale value of the vehicle as a percentage of the initial vehicle list price as the target variable and then train machine learning models such as random forests (Lessmann & Voß, 2017) and neural networks (Gleue et al., 2019) to make predictions of the residual value. The predictors used in the machine learning models comprise both the general vehicle

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configuration and the usage of the vehicle. However, to the best of our knowledge, none of the existing works have included granular equipment information in their models.

Vehicle equipment allows customers to tailor their cars according to their unique needs and preferences. When purchasing a new vehicle, customers have the choice to select their desired equipment. However, used vehicles come with a predefined set of equipment, which may not necessarily align with the preferences of the prospective buyer. For instance, a customer might not see value in a built-in navigation system if they prefer using their smartphone for navigation and, therefore, may not be willing to pay a premium for this equipment option. Thus, car dealers need to factor in the existing vehicle equipment when predicting the resale value of used vehicles, as it may be an important determinant of the final resale price. Yet, it is not straightforward to consider vehicle equipment information in machine learning predictions due to the vast array of equipment options and the resulting high-dimensional set of predictors.

Our work presents a machine learning prediction model for the resale price of used vehicles including equipment information as predictors. Based on a real-world dataset of 92,239 used car sales, we first perform feature engineering through a scalable, end-to-end automated procedure, wherein we preprocess 50,000 diverse equipment options presented as textual descriptions and then map them into 18 broad categories (e.g., navigation system, alloy rims). We find that including equipment information improves the prediction performance by 3.27%. This improvement is significant at common statistical significance thresholds. Finally, we account for the needs of our partner company that demands machine learning explainability in order to ensure reliable use in practice and, further, to identify profitable car configurations. Thereby, we offer insights that are relevant to real-world decision-making processes.

Our paper makes two main contributions. On the one hand, our work broadens the existing studies on used car resale price forecasting by handling granular equipment information. Specifically, we introduce an automated, end-to-end preprocessing approach for encoding high-dimensional equipment information, and, thus, our model offers a principled approach to improve resale price forecasts. On the other hand, our paper helps car retailers in generating more accurate resale price estimations, which may help them in making competitive offerings.

The remainder of the paper is organized as follows. Section 2 reviews earlier works on forecasting resale values of used cars. Section 3 presents our real-world dataset and introduces our machine learning approach with tailored preprocessing techniques to incorporate granular vehicle equipment. Section 4 provides the results from our prediction models where we assess the performance gain from the inclusion of granular vehicle equipment data. Finally, Section 5 discusses the managerial implications of our study for car leasing companies, while Section 6 concludes.

2. Related work

In this section, we first review relevant studies that focus on forecasting sales prices in general and, then, specifically within the context of used car sales. We continue by outlining how equipment data is currently utilized and, thereby, identify the use of granular equipment information as a salient gap in existing research, based on which we then motivate our study.

2.1. Models for predicting used car resale prices

Predicting future prices of sales plays a crucial role in successful retail businesses. It empowers retailers to mitigate potential future risks, allocate sufficient resources, adapt their product offerings as necessary, and is important for generating revenues. As a result, numerous studies have focused on predicting retail prices across various

industries, including wholesale trade (Escribano & Wang, 2021), real estate (Rapach & Strauss, 2009), and the financial industry (Guo et al., 2021). Our work, in contrast, focuses on predicting car prices. Table 1 provides a structured summary of previous research in this area, and we discuss the key dimensions of existing studies for predicting car prices in the following.

Initial studies primarily utilized hedonic pricing models, which aimed to mirror the process of car pricing over time (Cowling & Cubbin, 1972; Ohta & Griliches, 1976). Hedonic pricing models can thus provide valuable insights into how determinants of car prices change over time (Crawford & Neary, 2023; Mandys, 2023). However, hedonic pricing models fall short in producing accurate predictions due to the complex nonlinearities in how different car attributes relate to the final price. Furthermore, hedonic pricing models typically require manual feature engineering by choosing the functional form of the model or predictors (e.g., taking the logarithm for mileage) (Prado, 2009). Hence, hedonic pricing models are thus of little value for price predictions as needed nowadays by car leasing companies. Similarly building upon the components used in pricing cars, Sharma and Kumar Mitra (2024) present a multivariate adaptive regression spline model. They only use four car attributes and acknowledge that the interpretability of their model comes at the cost of reduced predictive performance compared to other machine learning models.

Du et al. (2009) present a genetic algorithm for optimal allocation of vehicles to different auction locations. For this, the authors use a linear pricing model. However, the authors encounter challenges in reliably estimating coefficients for certain car models and equipment segments, even with multiple years of sales history data. As a solution, they utilize the manufacturer's suggested retail price and a segment-specific depreciation curve to estimate each attribute's coefficient, which is a common practice in market research. However, such an approach requires manual and labor-intensive procedures. Importantly, much of the variation across car models and their specific configurations is lost. Simply adding up different attributes ignores any interactions and nonlinear relationships (e.g., Apple Car will be valued differently for an Audi than for a Seat).

In their study, Wu et al. (2009) construct an adaptive neuro-fuzzy inference system to forecast the resale prices of used cars. They evaluate their model using a manually-collected dataset comprising 192 used car sales. They find that by incorporating data from three specific equipment options and consolidating them into one predictor, the forecasting performance improves. Given that the model utilizes only four predictors in total, the authors suggest that this enhancement in predictive performance is likely due to the offsetting of other omitted variables, such as geographic differences and prevailing market conditions.

Recent studies aimed at used car sales prediction have predominantly utilized machine learning models due to their ability to handle nonlinear and complex relationships in the data. Lessmann and Voß (2017) examine a variety of prediction models and their corresponding performances. They find that nonlinear models, such as random forest ensembles, have strong predictive power. However, the authors randomly split the data into training and test sets, which is different from our work where we split the data chronologically to mirror a realistic decision-making process, ensuring that car sales in the training set precede those in the test set. Gleue et al. (2019) demonstrate that artificial neural networks achieve a strong prediction performance. The authors further analyze the role of different forecasting horizons, finding that prediction performance is lower for longer forecasting horizons. The authors also show that external factors, such as gasoline price levels, do not elevate prediction performance. Liu et al. (2022) present an optimized neural network model that incorporates feature selection and applies particle swarm optimization. They demonstrate the model's superiority on a dataset scraped at one point in time from an online platform offering used cars. Their prediction does not include

Table 1

Overview of related works on forecasting resale prices of used cars. To make accurate predictions, the dataset should be sufficiently large and include a suitable representation of the relevant technical vehicle specifications, sales details, and usage attributes. As only past sales can be used to forecast future sales, the data should be split chronologically (i.e., the hold-out set sales should be disjunct from the sales in the training set and chronologically later). By including equipment information, we later show that our model outperforms previous works.

Reference	Target variable	Dataset		Data split	Best performing model	Explainability	Equipment information
		Samples	Unique car models				
Crawford and Neary (2023)	Determinants for the price of new vehicles	3,160	36	–	Hedonic pricing model	✓	Coarse
Du et al. (2009)	Car re-location to maximize auction profits	345,000	Different Chrysler models	Chronologically	Nearest neighbor linear regression	✗	Coarse
Gleue et al. (2019)	Resale value	250,000	Different models of a German car manufacturer	Chronologically	Neural network	✓	Coarse
Lessmann and Voß (2017)	Resale value	454,618	6	Random	Ensemble selection, random forest	✗	Coarse
Liu et al. (2022)	Resale value	10,260	Representative of the Chinese automobile market	Random	Neural network	✗	–
Mandys (2023)	Determinants for the price of new vehicles	3,016	99% of the UK automobile market	–	Hedonic pricing model	✓	Coarse
Sharma and Kumar Mitra (2024)	Resale value	31,560	90% of the Indian automobile market	–	Multivariate adaptive regression splines	✓	–
Wu et al. (2009)	Resale value	192	4	No hold-out validation	Adaptive neuro-fuzzy inference system, neural network	✗	Coarse
<i>This paper</i>	Resale value	92,239	65	Chronologically	Gradient boosting	✓	Granular

any temporal component and, unlike our work, uses a very limited set of features to start with.

In light of the recent successes of gradient boosting in various business-oriented machine learning applications ([Lainder & Wolfinger, 2022](#); [Yi et al., 2023](#); [Zhang & Luo, 2023](#)), our research later utilizes LightGBM ([Ke et al., 2017](#)). However, to ensure robustness, we also experiment with other models for comparability (e.g., random forest, deep neural networks) ([Kraus et al., 2020](#)). One disadvantage of nonlinear models is typically their lack of transparency on how forecasts are derived. This may impede the trust of decision-makers in such models ([Ribeiro et al., 2016](#)), and, more importantly, it makes it challenging for decision-makers to identify profitable car model configurations. One study provides post-hoc explainability of the machine learning model by listing feature importance based on permutation importance ([Gleue et al., 2019](#)). While feature importance indicates the impact of an attribute on the prediction performance, it does not allow for any qualitative interpretation of the underlying relationships. To address this limitation, we thus make use of the SHAP value method ([Lundberg et al., 2017](#)) as a state-of-the-art method for machine learning explainability.

2.2. Input variables for predicting used car resale prices

Previous machine learning models for predicting resale prices make use of different inputs that act as predictors. Common predictors are general information on the car (e.g., brand, model), how it was used (e.g., mileage), and purchase details (e.g., the month and year of the sale).

However, predictors that categorize equipment are used only rarely. One work relies upon a manual annotation in which 3,016 textual descriptions of equipment options are manually processed and labeled ([Mandys, 2023](#)), yet such an approach is neither reproducible nor scalable in company practice. Another work ([Wu et al., 2009](#)) limits their manually scraped data set to just three equipment options, which are further condensed into a single predictive variable. Likewise, previous works utilizing machine learning techniques have a crucial limitation when it comes to equipment information. The reason is that equipment information enters the prediction model at a coarse level and not a granular level. For example, one work uses only the overall

special equipment price as a proxy ([Gleue et al., 2019](#)) while [Lessmann and Voß \(2017\)](#) use only a few “special” equipment options out of several thousand available equipment options. Hence, to the best of our knowledge, no work has so far assessed the predictive power of granular equipment information.

Including granular equipment information in machine learning comes with three main challenges. First, equipment data is high-dimensional. Per car model, manufacturers offer different configurations of vehicle parameters such as engine type and body style. These configurations can be further customized through different equipment options, which may be included by default or offered as optional add-ons. For instance, for every car model of the Volkswagen brand, a customer can select from an array of over 100 different equipment options and usually chooses around 40 to 50 of them. This thus gives a combinatorial space of different configurations in terms of vehicle equipment. Second, information on different equipment options only exists as descriptive texts. Such descriptive texts are not consistent across the brand but vary across car models (e.g., the description for a specific park assistance system may be different for a Volkswagen Polo and for a Volkswagen Passat), which adds an additional layer of complexity. Third, handling granular equipment options requires an effective computational representation. For example, one-hot encodings are the quasi-standard in machine learning ([James et al., 2013](#)), yet applying one-hot encodings directly to the raw equipment options is problematic for our study. The reason is that such one-hot encodings cannot generalize to new car models and would essentially require that, in practice, a machine learning model be updated with high frequency. Car models typically undergo updates at least annually and the descriptive texts often undergo updates alongside the model revisions. In our work, we address all of the three aforementioned challenges.

To the best of our knowledge, no prior research effectively utilizes granular equipment data to predict the resale value of used cars through a data-driven machine learning pipeline. Incorporating granular equipment data into machine learning models is challenging due to its high dimensionality. Importantly, our approach is closely aligned with practical needs in that it is applicable to new or changing equipment descriptions and further offers explainability to generate business-relevant insights that inform profitable car configurations.

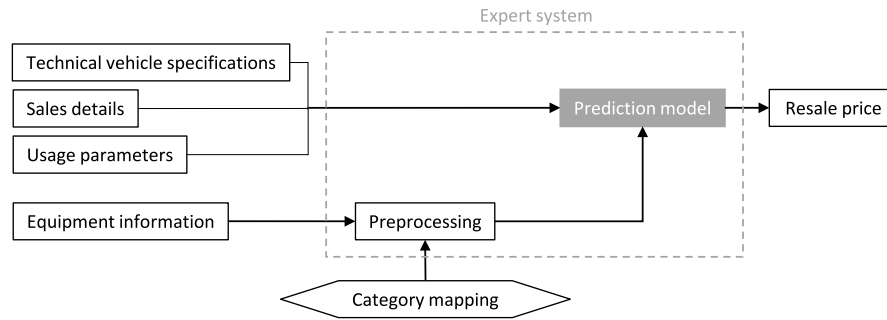


Fig. 1. Schematic illustration of a forecasting system for resale price prediction. This work focuses on the gain in prediction performance from using granular equipment data and integrating the summarized categories as predictors.

3. Method and materials

In this section, we introduce our machine learning model based on gradient boosting regression. We first give an overview of our real-world dataset of used car sales and detail our feature engineering to include equipment information in our prediction model. Finally, we provide implementation details. Fig. 1 provides an overview of our approach.

3.1. Car data

We evaluate our research question using a real-world dataset of used car sales provided by AMAG Group AG, the leading passenger vehicle retailer in Switzerland, with a market share of 31.4% (Hauri, 2022). AMAG operates in all geographic regions of the country, serving both the new and used car markets. The company deals exclusively with Volkswagen brands and offers customers a variety of acquisition models, including direct purchases and leasing options.

Our dataset consists of 92,239 used car sales transactions spanning over six years. Each transaction is characterized by a variety of attributes, such as mileage and horsepower. The dataset includes passenger cars from the brands Volkswagen (41%), Audi (31%), Skoda (20%), and Seat (8%). Collectively, these vehicles represent 65 different car models, including popular ones like Volkswagen Golf and Skoda Octavia. Interestingly, while six specific car models account for approximately 50% of the transactions in our dataset, there exist a long tail of less common car models, highlighting the diversity of the dataset. Table 2 provides a comprehensive overview of the attributes in our dataset.

During machine learning, we make predictions of our target variable y given by the relative resale value. Specifically, we use the ratio of the actual resale price to the new vehicle list price. The target variable y can thus be interpreted as the depreciation. Expressing resale prices as ratios is common in both practice and research (Lessmann & Voß, 2017; Prado, 2009). Using a ratio has several benefits in that it is directly comparable across both cheap and expensive cars and is thus also used by our partner company. In our results, we later ensure confidentiality of sensitive information by reporting the aforementioned ratios. Using the prediction $\hat{y}$, the residual value can then be computed by multiplying $\hat{y}$ with the original list price. As an example, consider a car leasing company that owns a car which was bought for USD 20,000. After the leasing contract has ended, the company sells it and recovers 10% of the original price of the acquisition. Therefore, the residual value is $10\% \cdot \text{USD } 20,000 = \text{USD } 2,000$. In our analysis, the resale price is based on actual transaction data recorded by AMAG while the list price is publicly accessible. Fig. 2 provides an overview of the distribution of the target variable y over time. Our dataset contains a broad range of depreciation values, with around half of them falling between 0.4 and 0.6.

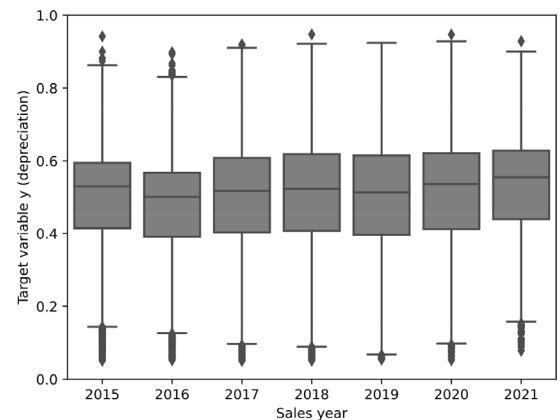


Fig. 2. Distribution of target variable y (depreciation) over time.

3.2. Equipment information

This paper focuses on the value of granular equipment information to improve forecasts of used car resale prices. Different equipment options allow customers to choose a car that best suits their needs. In general, a car, such as those our dataset, is typically described by around 50 different equipment attributes, each with a different categorical value (up to 50,000 equipment identifiers). Due to the high-dimensional nature of the different equipment attributes, we perform feature engineering through a data-driven, end-to-end procedure as follows.

First, we determine relevant categories of vehicle equipment (such as navigation or alloy rims). As our sales data is from transactions in Switzerland, we based our category selection on this market. We performed a pre-selection of relevant categories based on Autoscout24.ch¹ search terms. Autoscout24.ch allows users to filter for 50 equipment options. To make sure to only select the most differentiating ones, we validated these with 51 car dealers from our partner company. Based on the expert input, we narrowed our selection down to 18 target categories. While this step is manual, it simply generates a categorization that is meaningful and largely invariant across time (that is, there is no need to repeat the first step once new car models are brought onto the markets).

Our second step involves mapping the equipment of a specific car onto our predefined categories. In our dataset, we have close to 50,000 unique equipment identifiers in our dataset in textual form. Examples are “ZCA” (which stands for a specific navigation system) and “PH2”

¹ Autoscout24.ch is Switzerland’s biggest online car transaction platform (by annual transaction volume). The category selection is based on their extended search: <https://www.autoscout24.ch/de/auto/suche>.

Table 2
Overview of car data attributes.

Group	No. of attributes	Attribute names	Type
Technical vehicle specifications	11	Brand	Discrete
		Model year	Discrete
		Body type	Discrete
		Chassis size	Discrete
		Transmission	Discrete
		Fuel type	Discrete
		Drive train	Discrete
		Cubic capacity	Discrete
		No. of cylinders	Discrete
		Horsepower	Discrete
		Car dynamics	Discrete
Sales details	3	New vehicle list price	Continuous
		Year of initial registration	Discrete
		Year of used vehicle sale	Discrete
Usage parameters	2	Mileage	Continuous
		Age	Continuous
Granular equipment information	~50'000	ZCA, PH2, ... (= individual equipment identifiers)	Discrete
		No. serial equipment options	Discrete
		No. add-on options	Discrete
		Initial price of add-on options	Continuous

Table 3
Equipment description texts, English translations, and corresponding mapping onto equipment categories.

Original description [German]	Translation [English]	Equipment category
Navigationsfunktion "Discover media" fuer Radiosystem "Composition media"	Navigation function "discover media" for radio system "composition media"	Navigation system
Klimaanlage "Climatronic", 2-zonige Klimaanlage fuer Fahrer und Beifahrer	Air conditioning "Climatronic" 2-zone air conditioning for driver and front passenger	Air conditioning
Aluraeder 7,5j x 17 im 10-Speichen-Kristall Design, Reifen 225/50 r17	Aluminum wheel 7,5j x 17 in 10-spoke-crystal design, tires 225/50 r17	Alloy rims
Acoustic Park System vorne und hinten, mit Anzeige im MMI	Acoustic park system front and rear, with display in MMI	Park assist
Sitzbezüge in Stoff "Jacky" schwarz/rot (GTI)	Fabric seat covers "Jacky" black/red (GTI)	Seat coverings - fabric

(which stands for a specific air conditioning system). The equipment identifiers are represented by more than 10,000 different description texts. Description texts for the vehicle equipment are – for the most part – rather concise. Approximately half of them contain fewer than five words. Furthermore, description texts are only available in German, as illustrated by the examples and their respective English translations in Table 3. For the different keywords for each category, we then utilize a keyword-based search combined with regular expressions to decode the description texts and thus map a car's equipment onto the designated categories.

Our third step is to perform a one-hot encoding and thus yield a dataset with 18 one-hot-encoded predictors, that is, one per category. Each predictor thus indicates whether a car has equipment from the respective category (=1) or not (=0). Table 4 summarizes the categories by take rates in our dataset, showing that some equipment options are present in almost all cars (e.g., air conditioning) while others are found less than one out of four cars (e.g., head-up display, sunroof, electric seat adjustment).

The above approach was chosen for practical reasons. Upon deployment, our approach can be applied to cars in an automated manner and is thus scalable to large-scale datasets in practice. Furthermore, our approach is flexible as additional equipment can be easily accommodated in the future. We can readily add new car models with previously unseen description texts. For example, our mapping could be extended to incorporate new and emerging equipment technologies, thereby ensuring its relevance over time. Trends in equipment adoption can change rapidly. As technology advances and consumer preferences evolve, these shifts are likely to have a substantial impact on car resale values. One concrete example is the following: While in our dataset only ~20% of vehicles manufactured in 2017 were equipped with Android Auto/Apple CarPlay, this share increased to ~60% for

Table 4
Overview of equipment categories and corresponding take rate per equipment category in our dataset. To maintain confidentiality, take rates are summarized in ranges.

#	Equipment category	Take rate (observed)
1	Air conditioning	[75%–100%]
2	Alloy rims	[75%–100%]
3	Android Auto/Apple CarPlay	[25%–50%]
4	Cruise control	[75%–100%]
5	Electric seat adjustment	[0%–25%]
6	Electric tailgate	[0%–25%]
7	Head up display	[0%–25%]
8	Heated seats	[75%–100%]
9	Keyless access	[25%–50%]
10	Lane keep assist	[0%–25%]
11	Navigation system	[50%–75%]
12	Park assist	[75%–100%]
13	Rear view camera	[25%–50%]
14	Seat coverings – fabric	[25%–50%]
15	Seat coverings – leather	[25%–50%]
16	Sport seats	[25%–50%]
17	Sunroof	[0%–25%]
18	Trailer hitch	[0%–25%]

vehicles produced in 2020. Later, to ensure that our results are robust with respect temporal dynamics, we perform a sensitivity analysis in Section 4.3.

We later compare machine learning models that have access to granular equipment information with models that do not. Importantly, all models are trained using an otherwise identical, broad set of predictors consisting of technical vehicle specifications, sales details, and usage parameters. As a result, the only difference is whether we additionally input granular equipment information or not. Hence, all performance

gains must be attributed to the predictive power of granular equipment information.

3.3. Machine learning model

We use a gradient boosting regression (LightGBM; Ke et al., 2017) as our main prediction model. Gradient boosting has strong “out-of-the-box” performance and is less likely to overfit on the data as compared to other tree-based models (James et al., 2013). Gradient boosting is further able to model complex nonlinear relationships, performs well with high-dimensional data, and is fairly robust to outliers (Ke et al., 2017). The previous arguments are why we consider gradient boosting as highly relevant for our study: We are dealing with a dataset that encompasses a diverse array of automobiles, each uniquely defined by specific configuration and usage parameters. Additionally, given the real-world setting, our dataset includes vehicles with atypical pricing structures, necessitating a prediction model that can handle potential outliers. These reasons are also why gradient boosting has repeatedly been chosen in other settings with real-world data (Cui et al., 2018; Kadiyala et al., 2021; Xia et al., 2017).

A gradient boosting model consists of an ensemble of iteratively created regression trees (James et al., 2013). The first regression tree $\hat{f}^1(x)$ is fit to the true target value y , while subsequent regression trees $\hat{f}^b(x)$ are fit to the resulting prediction error, i.e., the residual r . Next, the residuals are updated via

$$r \leftarrow y - \lambda \hat{f}^b(x), \quad (1)$$

where λ is the learning rate. The final prediction model is a weighted ensemble of these individual decision trees, i.e.,

$$\hat{f}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x), \quad (2)$$

where B is the number of trees.

Later, in Section 4.5, we perform a series of robustness checks with alternative machine learning models (e.g., random forest, deep neural network) and with XGBoost as alternative gradient boosting approach to demonstrate that our choice yields the best prediction performance.

3.4. Implementation details

We perform 5-fold, fixed-size, rolling window cross-validation (Tashman, 2000) to account for the temporal order in our sales data. We keep a fixed-size window by pruning older observations to ensure that we only include more recent observations in the respective training sets. This allows our model to more closely capture the latest trends in both car resale prices and the changing popularity of vehicle equipment. Each training set contains at least one year of data. The hold-out set sales are always disjunct from the sales in the training set and are chronologically later. The reported results are the average performance across the five hold-out sets. For each subset, we tuned hyperparameters using Bayesian optimization via a tree-structured Parzen estimator approach implemented in the hyperopt package (Bergstra et al., 2011). Details on our hyperparameter tuning are in the Table A.11.

We predominantly report our model performance using the mean absolute error (MAE) as it provides an intuitive interpretation for decision-makers. We also report the root mean squared error (RMSE) in Section 4. Our key findings remain the same, independent of the metric. In our implementation, we set the loss function to mean squared error (MSE). This is a common choice for prediction models (Guo et al., 2021; Wasesa et al., 2017) with the advantage of being numerically differentiable.

4. Empirical results

This section reports our results from using machine learning for forecasting the resale price of used cars using granular equipment information. We first compare the performance of our machine learning model with and without the inclusion of granular equipment information. Subsequently, we examine how the prediction performance varies by car type and by the forecasting duration. We then proceed to understand the role of individual attributes using machine learning explainability. Finally, we perform a series of robustness checks.

4.1. Performance gain from vehicle equipment information

Table 5 reports the prediction performance. Specifically, we compare two model variants to assess the predictive power of granular vehicle equipment information: (1) We use a base model that uses all information excluding granular equipment information. (2) We compare the former against a model that additionally includes granular equipment information. This allows us to quantify the additional gain in predictive performance from using high-dimensional, granular equipment information. Both models are subject to rigorous hyperparameter tuning, so that all performance gains must be solely attributed to the use of granular equipment information.

Overall, we make the following observations. The base model achieves a MAE of 3.52%. It is outperformed by our model including granular equipment information, which yields an MAE of 3.40%. This corresponds to an improvement in prediction performance compared to the base model of 3.27%. A Wilcoxon signed-rank test confirms that the difference between the average MAEs among the base model and our model including granular equipment information is statistically significant at the 0.1% level (test statistic: 7.6×10^8 ; $p < 0.001$).

4.2. Prediction performance by car type

Different customers will have different preferences and willingness to pay for equipment in their vehicles. We therefore evaluate the prediction performance by different car types as equipment may be more or less informative for some car types. We thus split the vehicles in our dataset into three different clusters (using the k -means algorithm): compact, medium-sized, and premium cars. For each cluster, we trained a separate prediction model using the same implementation details as outlined in Section 3.4. Table 6 presents a breakdown of the prediction performance by different car types.

We find that performance gain from using equipment information is larger for compact cars than for medium-sized cars and premium cars. This is in line with literature aimed at understanding customer choice in the automotive sector, which finds that the influence of equipment on vehicle price diminishes as the price of the vehicles increases (Mandys, 2023). In our setting, the comparatively small performance gain for the premium car segment can be traced back to our partner company’s market positioning: the partner company mostly offers top-tier vehicles, which means that even their used cars are equipped with a rather comprehensive set of different vehicle equipment. As a result, certain equipment options are built into almost every car (e.g., almost all premium cars are equipped with alloy rims), so some of the equipment options are no longer discriminatory regarding resale price and thus do not have predictive power.

Notably, the prediction performance for the compact car model and the premium car model is lower than for the all-vehicle model. We attribute this to the comparatively smaller sample sizes in the clusters, with the largest cluster being less than half the size of the full dataset. A large sample size is especially important in our work due to the nature of our data in that granular vehicle information is typically sparse. Here, having a larger but diverse dataset appears to help representation learning by learning shared features across all clusters, which, in turn, finds a representation that is more effective in the downstream task. Similar observations were also made in other research but outside of the automotive industry (e.g., fraud detection Chen et al., 2022; image recognition Kang et al., 2011).

Table 5

Results comparing the prediction performance of the base model against our prediction model using equipment information. The results are reported as mean absolute error (MAE) and root mean squared error (RMSE) [lower = better]. The best-performing out-of-sample predictions are highlighted in bold. The absolute change refers to the absolute improvement over the base model, while the percentage change indicates the relative improvement.

Model	MAE (in %)		RMSE (in %)	
	In-sample (training set)	Out-of-sample (test set)	In-sample (training set)	Out-of-sample (test set)
(1) Base w/o equipment information	2.92	3.52	4.02	4.75
(2) Ours w/ equipment information	2.78	3.40	3.86	4.62
Absolute change	-0.14	-0.12	-0.16	-0.13
Relative change	-4.71%	-3.27%	-4.10%	-2.74%

Table 6

Prediction model performance across different car types in terms of MAE.

Model	MAE (in %) out-of-sample (test set)			
	All vehicles	Compact cars ($n = 42,585$)	Medium-sized cars ($n = 40,660$)	Premium cars ($n = 8,994$)
(1) Base w/o equipment information	3.52	3.75	3.41	3.48
(2) Ours w/ equipment information	3.40	3.60	3.37	3.46
Absolute change	-0.12	-0.15	-0.04	-0.02
Relative change	-3.27%	-3.99%	-1.14%	-0.43%

Table 7

Prediction model performance across different forecasting horizons in terms of MAE.

Model	MAE (in %) out-of-sample (test set)				
	10 months	17 months	23 months	28 months	32 months
(1) Base w/o equipment information	3.52	3.78	4.00	4.29	4.48
(2) Ours w/ equipment information	3.40	3.70	3.96	4.22	4.44
Absolute change	-0.12	-0.08	-0.04	-0.07	-0.04
Relative change	-3.27%	-2.20%	-1.05%	-1.47%	-0.95%

4.3. Sensitivity by forecasting horizon

The above results reported predictions for the current valuation of used cars. Here, we provide a sensitivity analysis where we predict resale prices several months ahead in the future. Specifically, we vary the forecasting horizons between 10 months and 32 months. Such analysis is relevant in car leasing where contracts are often made for one, two, or three years, and where the car leasing company may be interested in future returns from buy-back obligations.

We examine the prediction performance across different forecasting horizons. The results are in Table 7. Importantly, for all considered forecasting horizons, the inclusion of granular equipment information improves prediction performance. In general, the performance gain from equipment information is larger for shorter than for longer forecasting horizons. This can be expected and attributed to a concept drift in the underlying data (Zhang et al., 2011). That is, due to the longer forecasting horizon, the training data tends to become less representative of buyers' equipment preferences and willingness to pay in the hold-out dataset.

4.4. Machine learning explainability

Using machine learning explainability, we now seek to understand how our model arrives at predictions. We therefore identify the impact of specific attributes on the predicted values. In doing so, we generate insights relevant for management at our partner company by helping them understand what equipment options are found especially valuable by customers. This may inform our partner company when configuring new vehicles and eventually lead to an improved used car resale value and thus increase overall revenue. We infer customer valuations for different equipment options by employing the SHAP (SHapley Additive exPlanations) value method (Lundberg et al., 2017). The SHAP value method is a widely-used technique for interpreting opaque machine learning models such as gradient boosting (Molnar

et al., 2020; Senoner et al., 2022; Zhang & Luo, 2023). The SHAP value method generates, among others, a feature importance that measures the marginal influence of an individual predictor on the target variable. It thereby considers all possible permutations of the other predictors, thus explaining the extent to which the overall predictions are driven by some feature.

Fig. 3 provides an aggregated view of the SHAP values of all hold-out set cars. Both vehicle age and mileage exhibit the largest feature importance and, thus, the strongest predictive power. This is closely followed by other time-related attributes, which also contribute to our model's predictive strength. Examining the impact of equipment options, we see that emerging technologies, including Android Auto / Apple CarPlay and keyless entry, exert the most substantial positive effect on the predicted resale value. Conversely, customers of used cars appear unwilling to pay for features regarded as standard, such as air conditioning or cruise control.

4.5. Robustness checks

As part of a robustness check, we implemented additional machine learning models besides gradient boosting, namely, linear regression, ridge regression, lasso, a deep neural network, and a random forest. We also used XGBoost as an alternative gradient boosting approach. We performed rigorous hyperparameter tuning for all models according to best practice. The list of hyperparameters tuned for each model is available in the Table A.11. Detailed results are provided in Table 8. In line with Lessmann and Voß (2017), nonlinear models outperform linear models. The MAE and RMSE decreased for all models when including equipment information. Gradient boosting using the LightGBM implementation had the smallest error for the base model and when including equipment information. Altogether, this corroborates our main findings from above: vehicle equipment information provides significant predictive power.

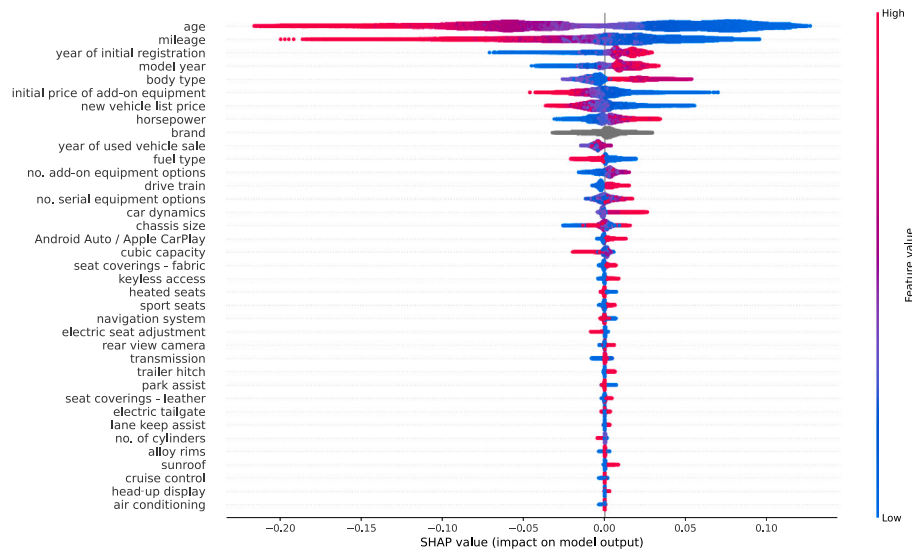


Fig. 3. SHAP summary plot showing the influence of individual equipment on the predicted resale price of cars.

Table 8

Prediction performance of different linear and nonlinear models. For all models, the prediction performance improves when including equipment information. Gradient boosting (LightGBM) yields the best performance in both cases, namely with and without including equipment information. The absolute change refers to the absolute improvement over the base model, while the percentage change indicates the relative improvement.

Model	Variant	MAE (in %)	RMSE (in %)
<i>Linear models</i>			
Linear regression	(1) Base w/o equipment information	4.50	5.88
	(2) Ours w/ equipment information	4.46	5.82
	Absolute change	-0.04	-0.06
	Relative change	-0.94%	-1.02%
Ridge regression	(1) Base w/o equipment information	4.50	5.88
	(2) Ours w/ equipment information	4.45	5.82
	Absolute change	-0.05	-0.06
	Relative change	-1.11%	-1.02%
Lasso	(1) Base w/o equipment information	4.50	5.88
	(2) Ours w/ equipment information	4.45	5.82
	Absolute change	-0.05	-0.06
	Relative change	-1.11%	-1.02%
<i>Nonlinear models</i>			
Deep neural network	(1) Base w/o equipment information	3.71	5.00
	(2) Ours w/ equipment information	3.52	4.77
	Absolute change	-0.19	-0.23
	Relative change	-5.12%	-4.60%
Random Forest	(1) Base w/o equipment information	3.89	5.16
	(2) Ours w/ equipment information	3.80	5.04
	Absolute change	-0.09	-0.12
	Relative change	-2.31%	-2.33%
Gradient boosting (XGBoost)	(1) Base w/o equipment information	3.86	5.12
	(2) Ours w/ equipment information	3.72	5.00
	Absolute change	-0.14	-0.12
	Relative change	-3.63%	-2.34%
Gradient boosting (LightGBM)	(1) Base w/o equipment information	3.52	4.75
	(2) Ours w/ equipment information	3.40	4.62
	Absolute change	-0.12	-0.13
	Relative change	-3.27%	-2.74%

We observe a slight increase in the target variable for 2020 and 2021 (see Fig. 2). This trend can be attributed to the impact of the COVID-19 pandemic and the subsequent global semiconductor shortage (Frieske & Stieler, 2022), which led to extended delivery times for new cars. As a result, there was an increased demand for used cars, driving up their value. To ensure the robustness of our model against these crisis-induced anomalies, we re-trained and evaluated

our model excluding data from the years 2020 and 2021. Our finding that granular equipment information improves prediction performance remains consistent even when these crisis years are removed. The detailed results are presented in Table 9.

We assess our approach using data from Switzerland, a country with diverse cultural contexts. To validate the robustness of our model across these differences, we split the test sets by language region.

Table 9

Prediction model performance excluding pandemic and semiconductor crisis years.

Model	MAE (in %) out-of-sample (test set)	
	Full dataset	Excluding 2020 and 2021
(1) Base w/o equipment information	3.52	3.48
(2) Ours w/ equipment information	3.40	3.41
Absolute change	−0.12	−0.07
Relative change	−3.27%	−2.01%

Table 10

Prediction model performance by Swiss regions.

Model	MAE (in %) out-of-sample (test set)			
	All regions	German-speaking	French-speaking	Italian-speaking
(1) Base w/o equipment information	3.52	3.45	3.48	3.65
(2) Ours w/ equipment information	3.40	3.32	3.36	3.56
Absolute change	−0.12	−0.12	−0.12	−0.10
Relative change	−3.27%	−3.57%	−3.45%	−2.66%

Table 10 reports prediction performance per geographic area. Our results demonstrate that the inclusion of granular equipment information consistently improves prediction performance across all regions.

Next to the individual equipment options, we included three attributes for better interpretability: The number of serial equipment options, the number of add-on equipment options and the money spent on add-on equipment options. While this information is typically easier to obtain and does not require extensive preprocessing, it might overlook some pertinent details. Simply extending the base model by these additional attributes improves the prediction performance by 2.77%, yet is inferior to granular equipment information. That is, integrating equipment identifiers resulted in an additional performance improvement of 0.50 percentage points. In sum, this corroborates our model choice from above in how we integrated granular vehicle equipment information.

Up until now, we have treated equipment categories as binary attributes. Such binary encoding only indicates whether a piece of equipment is present or absent. However, there are substantial quality variations within certain equipment categories. For instance, some navigation systems provide extra functionality compared to others, and larger alloy rims are typically seen as more valuable. To account for this, we integrated qualitative information about the equipment into our model. As an example, we included alloy rim size as a model attribute value. This modification did not result in a significant improvement in model performance. Similarly, to account for the concept drift mentioned in Section 4.3, we have tested encoding equipment by using the model year take rates and separately forecasted them for our hold-out dataset. However, when evaluating the final model performance, this did not yield a significant improvement in model performance either.

We finally evaluated the impact of including equipment specifications that are solely based on personal taste such as the color of the car's exterior or interior design. To ensure that our model could effectively manage the input, we adopted a similar strategy to our previous approach. Specifically, we categorized these attributes at a higher level, for instance, assigning both, a "slate gray" and a "charcoal gray" car simply "gray" as their color. However, including these taste-related attributes did not result in a significant improvement in model performance. This finding aligns with the experience of our partner company. They posited that cars with atypical or unique color combinations are not necessarily sold at a different price than cars with more mainstream colors. Instead, cars with unique colors tend to be simply sold more slowly.

5. Discussion

5.1. Implications for research

Our research contributes to research in the automotive retail sector and the broader field of data-driven prediction models. In particular, our machine learning model improves the prediction performance of resale price forecasts by incorporating granular vehicle equipment information. We find that our model decreases prediction errors (MAE) by 3.27%. When we compared our approach to the prediction errors reported in literature (Lessmann & Voß, 2017), we find that our approach yields a large improvement (14.34%), which we attribute to the use of gradient boosting together with granular equipment information.

Our machine learning model accommodates granular vehicle equipment information, which allows us to capture subtle variations between different vehicle configurations. To this end, our machine learning model must automatically process equipment identifiers in textual form. We achieve this through a keyword-based mapping. Although other papers have used text extraction to enhance prediction models (e.g., Abrahams et al., 2013; Bai et al., 2022; Feuerriegel et al., 2024; Liu et al., 2023; Maarouf et al., 2024), we are – to the best of our knowledge – the first to do so in the context of used car resale price prediction. This suggests possible benefits for other studies. Our approach can be transferred to domains such as pricing, fleet analysis, and market insights.

5.2. Implications for practice

Our work supports decision-makers in practice by informing them about how to set initial leasing prices and adjust their forecasts throughout the lease period. It allows car retailers to take preemptive measures, such as early buyouts, in case of discrepancies between the contracted resale price and the actual market price. The results from our analysis based on machine learning explainability (Section 4.4) provide additional insights at the individual equipment level. This in-depth understanding also enables decision-makers to encourage the configuration of vehicles that lead to favorable resale prices from the beginning, potentially through discount offers or marketing initiatives.

To demonstrate the impact for our partner company, we made a simple analysis and show that our machine learning model lowers the financial burden of the practice partner by approximately USD 455,970.21 annually. Here, we assume that the mean resale price of a car in our dataset is USD 26,500. Hence, reducing the MAE from 3.54% to 3.43% results in an absolute reduction of USD 2,933,200.20 over the entire dataset period, or USD 455,970.21 annually. Using a symmetric cost function, our model does not systematically over- or

Table A.11
Hyperparameters tuned for the different models.

Model	Hyperparameters tuned	Range
Linear regression	–	–
Ridge regression	Regularization strength α	$[1 \times 10^{-7}, 100]$
Lasso	Regularization strength α	$[1 \times 10^{-7}, 100]$
Deep neural network	Regularization strength α	$1 \times 10^{-5}; 0.01; 1; 100$
	Number of hidden layers	1; 2; 3
	Number of neurons in first hidden layer	10; 100
	Number of neurons in other hidden layers	10; 20; 100
Random Forest	Maximum tree depth	[2, 50]
	Number of trees	[15, 100]
Gradient Boosting (XGBoost)	Maximum tree depth	[2, 20]
	Number of trees	[15, 200]
	Learning rate	[0.01, 0.5]
	Feature fraction	[0.6, 1]
	Minimum child weight	[1, 10]
	Regularization strength α	[0, 1]
Gradient boosting (LightGBM)	Regularization strength λ	[0, 1]
	Maximum tree depth	[2, 20]
	Number of trees	[15, 200]
	Number of leaves	[20, 150]
	Learning rate	[0.01, 0.5]
	Feature fraction	[0.6, 1]
	Minimum number of data in one leaf	[20, 500]
	Regularization strength α	[0, 1]
	Regularization strength λ	[0, 1]

underestimate the resale price. Therefore, half can be attributed to resale prices that are set too high and half to resale prices that are set too low. Future work may study asymmetric cost functions (Dress et al., 2018); however, we abstain from this as it is not current practice at our partner company.

To deploy our prediction model in practice, we carefully address the needs of our partner company. In particular, we face challenges around the flexibility of the model design and the need for new business processes (Feuerriegel et al., 2022). To address the former, our machine learning approach was designed to offer great flexibility. For example, our preprocessing stage is automated end-to-end and is applicable to car models that were not part of the initial training set. This provides additional flexibility as requested by the business decision-makers of our partner company. To address the latter, our partner company revised its business decision-making process surrounding residual value estimation with the goal of minimizing prediction errors. As a result, they replaced the previous process based on expert judgments with one that is additionally based on a machine learning approach.

Once deployed, our machine learning prediction model should be subjected to regular updates. Practitioners should remember that the value of different equipment may vary over time (Crawford & Neary, 2023). For instance, built-in navigation systems, once the epitome of advanced features, are increasingly replaced by systems like Android Auto or Apple CarPlay. This necessitates regular re-evaluation of the relevant equipment and updating prediction models with the latest resale data. Furthermore, the predictive power of equipment information is likely to become smaller if all vehicles from a specific retailer (or within a particular segment) are equipped in a similar way. For instance, in our study, this holds true for the premium segment where we observe exceedingly high take rates, thereby minimizing the predictive power of equipment features. This highlights the need for an ongoing, detailed assessment of equipment take rates for individual fleets and each segment.

6. Conclusion

Our paper presents a machine learning model for improving the resale price forecasting of used vehicles by leveraging granular equipment information. To this end, we present an end-to-end feature engineering

to make equipment comparable across 65 different car models. Our keyword-based search allows for a straightforward extension of our prediction model to new equipment descriptions and newly emerging equipment categories. We find that including equipment categories in our prediction model improves the prediction performance (MAE) by 3.27%. We further use machine learning explainability to determine the impact of individual options on the predicted resale value, which generates managerially-relevant insights. Specifically, it allows business decision-makers to create resale value-optimal configurations and thus presents the prospect of overall larger resale values.

CRediT authorship contribution statement

Svenja Bergmann: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing, Visualization. **Stefan Feuerriegel:** Conceptualization, Methodology, Writing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Svenja Bergmann reports a relationship with AMAG Group AG that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. AMAG Group AG had no influence on the design, analysis or interpretation of the results.

Appendix. Hyperparameters tuned for the different models

See Table A.11.

Data availability

The authors do not have permission to share data.

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